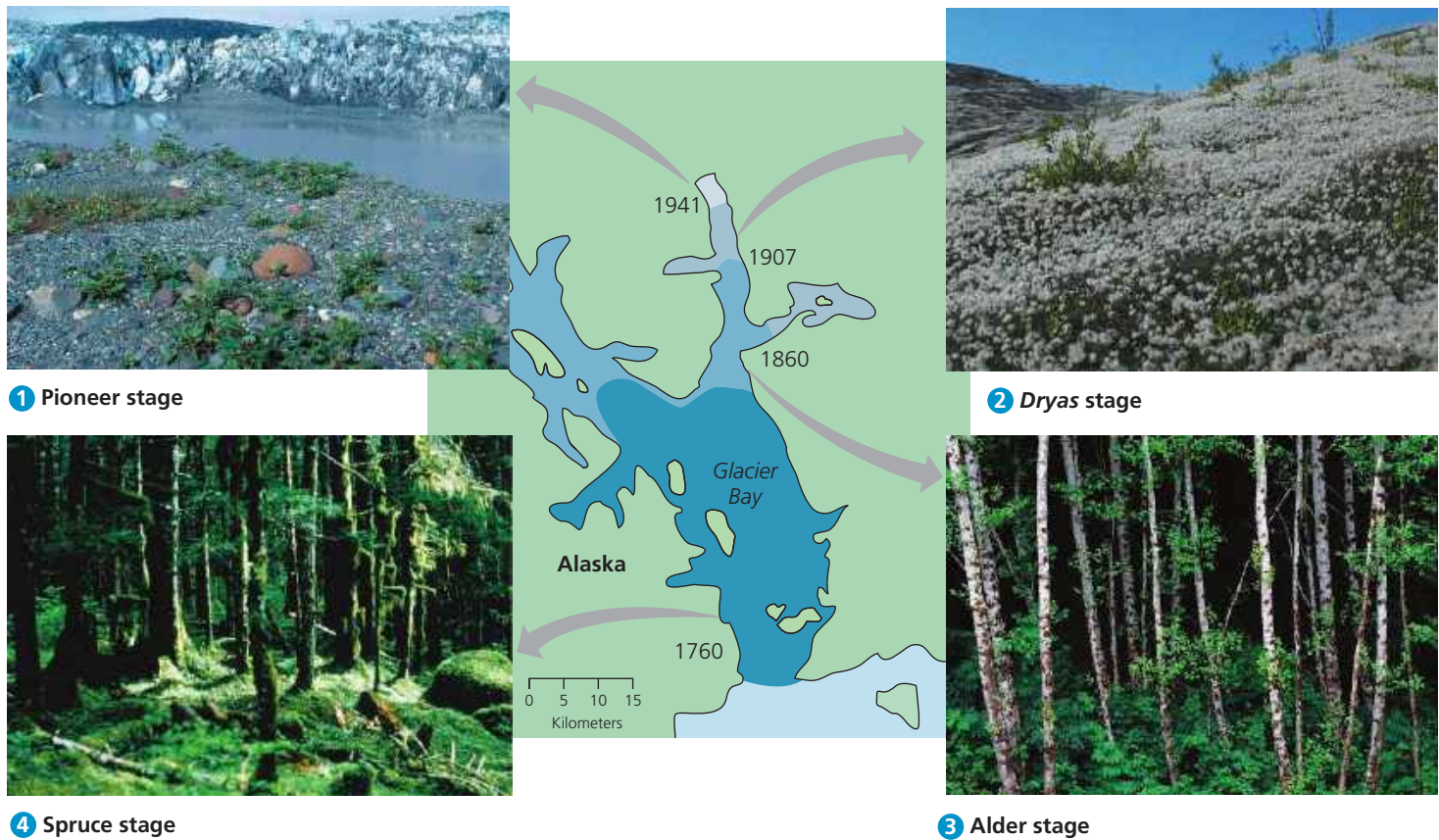




▲ **Figure 54.26** Glacial retreat and primary succession near the shores of Glacier Bay, Alaska. The different shades of blue on the map are based on historical information of how the area covered by Glacier Bay has increased over time due to retreat of the glaciers since 1760.

Producing such a community through primary succession may take hundreds or thousands of years.

Early-arriving species and later-arriving ones may be linked by one of three key processes. The early arrivals may *facilitate* the appearance of the later species by making the environment more favorable—for example, by increasing the fertility of the soil. Alternatively, the early species may *inhibit* establishment of the later species, so that successful colonization by later species occurs in spite of, rather than because of, the activities of the early species. Finally, the early species may be completely independent of the later species, which *tolerate* conditions created early in succession but are neither helped nor hindered by early species.

Ecologists have conducted some of the most extensive research on primary succession at Glacier Bay in southeastern Alaska, where glaciers have retreated more than 100 km since 1760 (Figure 54.26). By studying the communities at different distances from the mouth of the bay—with the oldest exposed areas being closest to the mouth and the more recently exposed areas being farthest—ecologists can examine different stages in succession. As the glaciers have retreated, the following sequence of events occurred on areas above the waterline of the bay: ① The exposed glacial moraine is colonized first by pioneering species that include

liverworts, mosses, fireweed, scattered *Dryas* (a mat-forming shrub), and willows. ② After about three decades, *Dryas* dominates the plant community. ③ A few decades later, the area is invaded by alder, which forms dense thickets up to 9 m tall. ④ In the next two centuries, these alder stands are overgrown first by Sitka spruce and later by western hemlock and mountain hemlock. In areas of poor drainage, the forest floor of this spruce-hemlock forest is invaded by sphagnum moss, which holds water and acidifies the soil, eventually killing the trees. Thus, by about 300 years after glacial retreat, the vegetation consists of sphagnum bogs on the poorly drained flat areas and spruce-hemlock forest on the well-drained slopes.

Succession on glacial moraines is related to changes in soil nutrients and other environmental factors caused by transitions in the vegetation. Because the bare soil after glacial retreat is low in nitrogen content, almost all the pioneer plant species begin succession with poor growth and yellow leaves due to limited nitrogen supply. The exceptions are *Dryas* and alder, whose roots host symbiotic bacteria that fix atmospheric nitrogen (see Figure 37.12). Soil nitrogen content increases quickly during the alder stage of succession and keeps increasing during the spruce stage (Figure 54.27). By altering soil properties, pioneer plant